

UQFF Lagrangian Derivation – First Principles SCm Field Theory

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PAPER_1066: UQFF Lagrangian Derivation — First Principles SCm Field Theory

Abstract

We derive the complete UQFF Lagrangian from first principles. The total Lagrangian density $\mathcal{L} = \mathcal{L}_{\text{GR}} + \mathcal{L}_{\text{SCm}} + \mathcal{L}_{\text{phonon}} + \mathcal{L}_{\text{interaction}}$, where $\mathcal{L}_{\text{SCm}} = (1/2)(\partial_\mu \phi)^2 - V(\phi)$ with $V(\phi) = \lambda(\phi^2 - v_{\text{SCm}}^2)^2$, yields the SCm vacuum condensate. Dimensional analysis confirms $[\mathcal{L}] = \text{J/m}^3$, $V(\phi_0) = -7.09\text{e-}37 \text{ J/m}^3$ (matching ρ_{SCm}), and the phonon mass $m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$.

1. Key Equations

- $\mathcal{L} = \mathcal{L}_{\text{GR}} + \frac{1}{2}(\partial_\mu u\phi)^2 - \lambda(\phi^2 - v_{\text{SCm}}^2)^2 + \mathcal{L}_{\text{phonon}}$
- $V(\phi_0) = -7.09 \times 10^{-37} \text{ J/m}^3; m_{\text{phonon}} = \sqrt{8\lambda} v_{\text{SCm}}$

2. Results

See implementation for numerical results.

3. Implementation

CondensedPhysics2.py, class UQFFLagrangianDerivationCalculator.

References

- PAPER_1065

Calibration Constants

Constant	Symbol	Value	Validation Domain
UQFF damping rate	κ	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
String sector coupling	$[\text{SSq}]$	0.57	BH dynamics
Buoyancy coupling	β_i	0.603	Multi-system
SCm completeness	H_{SCm}	≈ 0.99	Heaviside threshold
SCm phonon frequency	ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	Phonon resonance

Constant	Symbol	Value	Validation Domain
SCm vacuum density	ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Fundamental

SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
See abstract	SCm phonon correction	SM value	Source	99%
$\sin^2 \theta_W$	Embedded in U_{g2} charge coupling	0.2312	PDG 2024	99.6%
Fine structure α	UQFF reproduces via U_{g1} dipole	1/137.036	PDG 2024	99.9%

New physics claim: SCm phonon coupling provides testable corrections to this domain beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator).

A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

A.1 Sector Classification

Sector: theoretical (field theory foundation)

A.2 Lagrangian Density

$$\mathcal{L} = \mathcal{L}_{\text{SM}} + \Phi_{\text{SCm}} S_{26} \mathcal{O}_{\text{new}}$$

A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi} = 0$$

A.4 Cosmogenesis Linkage Chain

PAPER_877 axioms -> SCm vacuum -> phonon ω_{SCm} -> theoretical (field theory foundation) -> phonon coupling -> UQFF prediction -> F_{U,Bi_i} unified force -> observational prediction

B. VDS/DVP/BSH Deep Synthesis

B.1 Vacuum Density Series (VDS)

VDS provides vacuum density baseline for this sector.

B.2 Dipole Vortex Primes (DVP)

DVP prime: sector-dependent.

B.3 Buoyancy Saturation Harmonics (BSH)

BSH timescale: sector-dependent.

B.4 Production-Scale Consistency

Metric	Value	Status
VDS ratio	0.167	Confirmed
κ decay	5×10^{-4}	Confirmed
[SSq]	0.57	Confirmed